

COS30002 AI for Games – Spike Extension Report

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Contents

Executive Summary.....	2
Unit's Learning Outcomes	2
Extension Alignment Matrix.....	3
Extension Alignment Justification.....	4
Task 3: Tic-Tac-Toe AI Battles — ULO 5	4
Task 4: Perfect AI (Minimax) — ULO 4	4
Task 4: Tree Visualisation — ULO 1, 4	4
Task 5: Custom Text Map Files — ULO 2	4
Task 5: New Terrain Options — ULO 1, 2	4
Task 5: Animated Agent — ULO 2	5
Task 6: Non-Square Tessellations — ULO 1, 2	5
Task 9: Advanced Tactical AI — ULO 4	5
Task 9: Spatial Distance Tracking — ULO 4	5
Task 10: Symmetry Analysis — ULO 1, 5	5
Task 11: Agent Pursuit — ULO 3, 4.....	5
Task 13: Obstacle Avoidance — ULO 3, 4	5
Task 13: Hunter State Transitions — ULO 1, 4, 5.....	6
Task 14: Non-Overlapping Agents — ULO 3, 4.....	6
Task 14: Predator Avoidance — ULO 3, 4	6
Task 14: Wall Avoidance — ULO 3, 4	6
Task 15: Fire Rate and Range — ULO 1, 4.....	6
Task 16: Extended Codebase — ULO 4, 5	6
Conclusion	7

Executive Summary

The purpose of this report is to build a documented summary of all the extensions that I have done over the course of the unit and display how they adhere to the unit's learning outcomes. With this report, teaching staff will be able to identify what extensions to my spike work I have done, ideally lending me a strong position when my overall work is being assessed.

Unit's Learning Outcomes

- ULO1 Discuss and implement software development techniques to support the creation of AI behaviour in games, focusing on non-obvious problem-solving approaches.
- ULO2 Understand and utilise a variety of graph and path-planning techniques.
- ULO3 Design and create realistic movements for agents using steering force models to simulate natural and responsive behaviour.
- ULO4 Design and develop agents capable of planning actions that require adaptive problem-solving techniques and innovative solutions.
- ULO5 Combine multiple AI techniques to develop sophisticated game AI capable of solving intricate, multi-layered problems that arise in dynamic game environments.

Extension Alignment Matrix

Extension	ULO 1	ULO 2	ULO 3	ULO 4	ULO 5
Task 3: Tic-Tac-Toe AI Battles					X
Task 4: Perfect AI				X	
Task 4: Tree Visualisation	X			X	
Task 5: Custom Text Map Files		X			
Task 5: New Terrain Options	X	X			
Task 5: Animated Agent		X			
Task 6: Non-Square Tessellations	X	X			
Task 9: Advanced Tactical AI				X	
Task 9: Spatial Distance Tracking				X	
Task 10: Symmetry Analysis	X				X
Task 11: Agent Pursuit			X	X	
Task 13: Obstacle Avoidance			X	X	
Task 13: Hunter State Transitions	X			X	X
Task 14: Non-Overlapping Agents			X	X	
Task 14: Predator Avoidance			X	X	
Task 14: Wall Avoidance			X	X	
Task 15: Fire Rate and Range	X			X	
Task 16: Extended Codebase				X	X

Extension Alignment Justification

Task 3: Tic-Tac-Toe AI Battles — ULO 5

This extension demonstrates the development of a controlled experimental environment in which two autonomous agents are evaluated through repeated interaction. The emphasis is on combining AI decision-making with structured simulation to observe emergent behaviour across multiple iterations. This aligns with ULO 5 as it reflects the integration of multiple AI components (agents, game loop, and evaluation structure) to produce a system capable of resolving a dynamic, multi-agent problem space.

Task 4: Perfect AI (Minimax) — ULO 4

This implementation represents a fully deterministic decision-making agent capable of exhaustively evaluating future game states to guarantee optimal outcomes. It aligns with ULO 4 as it demonstrates adaptive planning through systematic evaluation of possible actions and outcomes, producing a robust solution to adversarial decision-making problems in a constrained environment.

Task 4: Tree Visualisation — ULO 1, 4

The tree visualisation provides interpretability of the AI's decision-making process by exposing the evaluated search space. This aligns with ULO 4 through its representation of structured planning behaviour, and with ULO 1 as it reflects a software development practice that supports debugging, validation, and understanding of non-obvious AI behaviour within a game system.

Task 5: Custom Text Map Files — ULO 2

This extension supports the representation of navigable environments through externally defined spatial data structures. It aligns with ULO 2 as it enables the application of graph-based pathfinding techniques over user-defined map configurations, reinforcing understanding of structured spatial problem-solving.

Task 5: New Terrain Options — ULO 1, 2

The inclusion of terrain-dependent cost variation demonstrates how pathfinding behaviour adapts to changing environmental constraints. This aligns with ULO 2 through the application of weighted graph traversal techniques such as A* and Dijkstra, and with ULO 1 as it reflects the integration of software systems that enable more complex and realistic AI behaviour modelling.

Task 5: Animated Agent — ULO 2

The animated traversal of computed paths represents the translation of abstract pathfinding results into observable movement behaviour. This aligns with ULO 2 as it demonstrates the practical application of graph-based route computation within a simulated environment.

Task 6: Non-Square Tessellations — ULO 1, 2

This extension demonstrates the adaptation of classical pathfinding techniques to alternative spatial meshes. It aligns with ULO 2 through the extension of graph traversal algorithms to non-standard geometries, and with ULO 1 as it reflects the engineering effort required to adapt heuristic reasoning and neighbour relationships to novel spatial representations.

Task 9: Advanced Tactical AI — ULO 4

This work demonstrates agents capable of interpreting complex game states and selecting actions based on situational awareness rather than static rules. It aligns with ULO 4 as it embodies adaptive decision-making under uncertainty, requiring the agent to evaluate and respond to dynamic conditions.

Task 9: Spatial Distance Tracking — ULO 4

This extension reflects agents that incorporate real-time environmental information such as spatial proximity and threat detection into their decision-making processes. It aligns with ULO 4 as it demonstrates adaptive behavioural responses based on evolving game state information.

Task 10: Symmetry Analysis — ULO 1, 5

This analytical component evaluates agent performance across different map configurations using empirical reasoning. It aligns with ULO 5 through the synthesis of multiple performance factors and comparative evaluation, and with ULO 1 as it reflects critical analysis of AI behaviour using structured evidence and statistical reasoning.

Task 11: Agent Pursuit — ULO 3, 4

This extension demonstrates predictive movement behaviour where agents anticipate future target positions rather than reacting to current states. It aligns with ULO 3 through the implementation of steering-based movement modelling, and with ULO 4 through adaptive prediction and decision-making in a dynamic environment.

Task 13: Obstacle Avoidance — ULO 3, 4

This behaviour demonstrates agents capable of navigating environments while maintaining separation from obstacles through local steering responses. It aligns with

ULO 3 as it models realistic movement dynamics, and with ULO 4 as it requires continuous adaptation to environmental constraints during navigation.

Task 13: Hunter State Transitions — ULO 1, 4, 5

This extension demonstrates an agent operating under multiple behavioural states, transitioning between exploration and pursuit based on environmental conditions. It aligns with ULO 4 through adaptive decision-making, with ULO 5 through the integration of state-based logic and behavioural systems, and with ULO 1 through the structured implementation of a multi-state AI architecture.

Task 14: Non-Overlapping Agents — ULO 3, 4

This work ensures physical separation between agents through collision constraints and behavioural adjustment. It aligns with ULO 3 through realistic movement simulation and spatial interaction modelling, and with ULO 4 through adaptive response to spatial constraints in a dynamic environment.

Task 14: Predator Avoidance — ULO 3, 4

This extension demonstrates reactive agent behaviour in response to perceived threats within the simulation environment. It aligns with ULO 3 through steering-based avoidance mechanics and with ULO 4 through adaptive decision-making based on environmental stimuli.

Task 14: Wall Avoidance — ULO 3, 4

This behaviour models agents that maintain spatial awareness of environmental boundaries and adjust movement accordingly. It aligns with ULO 3 through realistic steering behaviour implementation and with ULO 4 through continuous adaptive response to static obstacles in the environment.

Task 15: Fire Rate and Range — ULO 1, 4

This extension incorporates weapon constraints into the agent's decision-making process, requiring consideration of optimal engagement conditions. It aligns with ULO 4 through adaptive combat decision-making based on situational variables, and with ULO 1 through integration of multiple gameplay systems influencing AI behaviour.

Task 16: Extended Codebase — ULO 4, 5

This extension demonstrates a cohesive simulation system in which multiple AI behaviours and environmental interactions operate simultaneously without destabilising core logic. It aligns with ULO 4 through adaptive agent decision-making within a complex environment, and with ULO 5 through the integration of multiple interacting AI systems into a unified operational framework.

Conclusion

The key outcomes of this report, such as the alignment matrix and alignment justification, all serve to communicate my understanding of the unit's core concepts. The purpose of these outcomes is to illustrate the way my implementation of these concepts has evolved through the manner in which I have applied them over the course of the unit.